Patrick Prell
Robotics & Embedded Systems Engineer

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I grew up on a sheep farm in the Southern Tablelands. For eight years I've built hardware and software that ships to real customers. I started on ArduPilot and PX4 survey UAVs at CORDEL. That work covered bring-up, control tuning, mission planning and field flight operations. I also built an EKF that fused IMU, RTK GNSS and LiDAR to localise a survey platform outdoors.

At ResusRight I led firmware for a clinical monitor under IEC 62304 and ISO 13485, where reliability was not negotiable. I now build RF direction-finding systems and embedded Linux at DroneShield, on products that deploy to the field. I own systems end to end and mentor the engineers around me.

Skills

Aerial Robotics	**ArduPilot and PX4** flight stacks, UAV bring-up, tuning and mission planning, GNSS/RTK navigation, field flight operations
Control & Estimation	**Extended Kalman Filter, Bayesian sensor fusion**, full-state feedback, observer design, fixed-rate discrete-time control loops, MATLAB
Robotics & Autonomy	**Localisation and pose estimation (IMU / RTK GNSS / LiDAR)**, vision-based navigation, point cloud processing, ROS 2 (working knowledge)
Perception & Vision	**OpenCV (C++)**, optical flow, 3D reconstruction from 2D imagery, calibration, ground-truth evaluation and visualisation
Embedded & Systems	**Embedded Linux**, bare-metal C/C++, bootloaders, STM32 / NXP / Silicon Labs / ESP32, Bluetooth LE, gRPC, Qt, Flutter, JavaScript, POSIX shell
Field & Hardware	PCB design (KiCad, Eagle), 3D CAD (PTC Creo, Fusion 360), SMD soldering and rework, 3D printing (FDM/SLA), CNC – comfortable taking hardware from prototype to production
Leadership	Led small engineering teams to on-time delivery, mentoring and university tutoring, coordinating work across production, hardware and test engineering

Professional Experience

Current – Nov 2023
Embedded Software Engineer | DroneShield - Australia
- Delivered angle-of-arrival estimation and the conducted calibration behind it for a four-antenna direction-finding system (400 MHz–7 GHz). Designed simultaneous multi-channel injection through a characterised splitter, producing a per-channel correction table that removes amplitude and phase imbalance and makes bearing accuracy repeatable unit to unit.
- Cut analog front-end calibration from over 12 hours to 40 minutes – a **20x** improvement that removed the single largest bottleneck in production. Required a full rearchitecture and coordination across production, hardware and test engineering.
- Built Python (Dash/Plotly) analysis tooling for frequency sweeps and calibration results. Stitched S2P measurements across 11 filter banks into a single validated front-end response.

Embedded C++ · Python · Rust · RF Calibration · Direction Finding · FPGA/RFSoC · Qt · gRPC · Embedded Linux · numpy/scipy · Dash/Plotly · POSIX Shell · Jira · GitLab

May 2025 – Sep 2019
Co-founder & CTO | Borne Clothing - Australia
- Co-founded and grew a social-enterprise clothing business to a successful exit. Donated $30,000+ to the Against Malaria Foundation – 5,000 families protected from mosquito-borne disease for three years.

Business Operations · Web Development · Lean Growth

Sep 2023 – May 2022
Embedded Systems Engineer | ResusRight - Australia
- **Led firmware development of a manufacturing test jig**, managing a small team and setting timelines that delivered on schedule and unblocked board production.
- Architected a Qt application on an embedded Linux SOM, including system architecture and the BLE stack. Developed firmware for "Nemo", a clinical resuscitation monitor now used on real patients (NXP and Silicon Labs MCUs, BLE and serial stacks).
- Took over "Juno" firmware (ESP32, BLE) and designed its Flutter companion app with Firebase auth/database and iOS/Android Bluetooth stacks.
- Worked to IEC 62304, IEC 60601 and ISO 13485 – safety-critical, traceable firmware in a startup moving at pace.

Embedded C/C++ · Embedded Linux · Qt · NXP μC · Silicon Labs μC · Espressif μC · Bluetooth LE · IEC62304 · IEC60601 · ISO13485 · Flutter · Firebase · Python · GitHub

May 2022 – Apr 2018 | **Mechatronic Engineer** | CORDEL (formerly Airsight) - Australia

- **Worked hands-on with ArduPilot and PX4 flight stacks** across the company's survey UAV fleet – airframe bring-up, flight controller configuration and tuning, mission planning and field flight operations.
- Contributed to **GNSS navigation research**, characterising RTK positioning accuracy and the behaviour of GNSS-based navigation under real field conditions.
- **Implemented an Extended Kalman Filter fusing IMU, RTK GNSS and LiDAR** for pose estimation and localisation of a survey platform in unstructured outdoor environments. Required Bayesian filtering, linear algebra and careful handling of dropout and noise in real field data.
- Developed pose and velocity estimation from optical flow in a C++/OpenCV pipeline. Spent a year in New York working largely unsupervised on 3D point extraction from 2D camera footage on a moving train.
- Implemented a pseudo-RTOS to arbitrate commands over the serial port and designed the status indicator board – PCB, C and AVR assembly firmware.

ArduPilot | PX4 | UAV Flight Operations | GNSS/RTK | Mission Planning | C++ | Python | OpenCV | Extended Kalman Filter | Sensor Fusion | LiDAR | RTK GNSS | Point Clouds | Embedded C | AVR Assembly | RTOS | KiCad | Rust | Jira | Bitbucket

Projects

Farm inspection drone in ROS 2 / PX4 SITL — 2026

github.com/Prelly95/paddock

Autonomous ROS 2 inspection stack for a VTOL platform, built to help my family monitor our farm. Autonomous take-off/land with fixed-wing cruise between paddocks. Missions gated on live flight conditions inferred from local weather reports. Telemetry feeds a homestead C2 dashboard for water level and asset monitoring.

ROS 2 | rclcpp | PX4 | uXRCE-DDS | Gazebo | EKF | Guidance & Control | C++ | Python

UVie – magnetic UV exposure tracker — 2026 – Current

github.com/Prelly95/uvie-hardware , github.com/Prelly95/uvie-firmware , github.com/Prelly95/uvie-app

Battery-free wearable: nRF54L15, LTR390-UV-01 sensor, solar harvesting (BQ25570 + supercapacitor). Embedded Rust firmware with BLE/NFC. SolidJS + Capacitor companion app sharing a no_std Rust wire protocol via WebAssembly. Hardware to app.

Embedded Rust | nRF54L15 | KiCad | BLE | NFC | Energy Harvesting | SolidJS | Capacitor | WebAssembly

Two-wheeled balancing robot — 2017

github.com/Prelly95/MCHA3K

End-to-end build – PCB, plant model, Kalman observer and full-state feedback controller in MATLAB, implemented at 50 Hz in embedded C on an 8-bit ATMega32 (no FPU).

Full-State Feedback | Kalman Observer | Real-Time Control | Embedded C | PCB Design | MATLAB | Eagle

Education

Jul 2019 – Feb 2015 | **B.Eng. Mechatronics (Honours)** | University of Newcastle

Feb 2022 – Jan 2022 | **Vision-Based Navigation** | Industry Short Course - Dr Chris Renton

References

Available on request.